

SMART CONTRACT SECURITY AUDIT - HOTELBOOKING

One-Page Visual Audit Report

Purpose: Hotel registry and room booking system using ETH payments.

Risk	Severity	SWC	OWASP
bookHotel() creates a booking but never stores it	Critical	N/A	SC02 Business Logic
Users pay ETH but receive no persistent reservation record	High	SWC-132	SC02 Financial Risks
Contract collects ETH with no owner withdrawal or refund path	Medium	SWC-132	SC02 Financial Risks
getCustomerBookings() performs unbounded loops over bookings	Medium	SWC-128	SC02 Gas/DoS

Root Cause: The core booking flow is incomplete: the Booking struct is created in memory but not pushed into hotel storage.

Business Logic Findings: Customers can pay for a booking that is never recorded. Funds can accumulate without operational settlement, refunds, or hotel payout logic.

BENCHMARK METRICS

Time_seconds: 75 | Confidence: 0.96 | Total_findings: 4

Critical: 1 | High: 1 | Medium: 2 | Low: 0

Precision_estimate: 0.94 | Coverage_estimate: 1.00 | Risk_score: 86/100

GOLD STANDARD EVALUATION

TP: N/A | FP: N/A | FN: N/A | Precision: N/A | Recall: N/A | F1: N/A